REAL TIME DROWSINESS DETECTION USING OPENCV

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***Abstract*— Driver's sleepiness is a leading cause of crashes, and sometimes even leads to fatalities. In this paper, we propose an effective and real-time drowsiness detection framework to track and analyze the important facial indicators with the help of state-of-the-art computer vision and machine learning algorithms. It uses the OpenCV and the Dlib libraries for fast reliable facial landmark detections and tracking, allowing to monitor three relevant physical drowsiness markers, namely, eye closure, yawning, and head position. Eye Aspect Ratio (EAR) is calculated to recognize the blink period of eye, whereas Mouth Aspect Ratio (MAR) is employed to recognize yawn by detecting elongated mouth openings. Head pose estimation is also used to track attentiveness and detect nodding or head tilting as signs of sleepiness. The system runs in real-time and can accommodate changes in illumination, face poses, and driver appearance. Experimental studies show that its accuracy and reliability are quite high under different conditions, which indicates that it is a feasible and cost-effective non-invasive method for an ITS. The proposed model can shed light on the design of Advanced driver assistance system (ADAS) to bring down fatigue related crashes and thus to improve road safety.**

Index Terms---- Opencv, Dlib Libraries, Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Facial Landmarks, advanced driver assistance systems (ADAS).

INTRODUCTION

The road safety is still a serious public health issue, driver fatigue and drowsiness are major factors in traffic accidents and deaths in the world. In the face of this increasing problem intelligent driver assistance systems (IDAS) have become a necessity for driver alert monitoring and avoiding road accidents [2]. Of these, vision-based methods using computer vision techniques and machine learning have shown promising performance for real-time tracking and analysis of a driver [1].

In this paper, a robust non-intrusive driver drowsiness and head posture system is proposed based on image and facial landmark detection. The implemented solution uses the OpenCV and Dlib in order to capture facial landmarks then calculate the Eye Aspect Ratio (EAR) as well as monitor head movements which are all used to measure signs of fatigue or inattention. With the combination of these parts and real-time video processing, the system can inform the drivers when it comes to critical moment such as fatigue or driving lack, enhance road security, and decrease the accident rate [9]. Drowsy driving detection systems are systems designed to detect indication of fatigue will by observing different physiological and behavioral signs. Although traditional methods (i.e. EEG, EOG, and HRM) are accurate, they rely on intrusive hardware and are not easily implemented as in-car systems. On the other hand, computer vision-based approaches have been proposed as a non-intrusive and economic solution making use of facial analysis methods to track eye closure, blink rate, yawn and head position [16]. The primary objective of the system is to maintain the constant state of the driver's facial appearance without physical contact or wearable sensors and use lightweight algorithms optimized for real-time processing of the data so that there is minimal latency in detection of symptoms of drowsiness and when detected the system can issue auditory or visual alerts to re-engage the driver and prevent the driver from committing an accident[15].This study fills the void between traditional driver monitoring approaches and smart dynamic fatigue detection systems. The performance evaluation is described in the following sections to justify the effectiveness and robustness of the system in practical environments [10].

In Summary The main novelty of this work is that it combines multiple facial behavior metrics such as eye closure, yawning, and head position into one simple, unified detection framework that can be performed in real time without the use of specialized hardware. The system is highly adapted to different lighting conditions and orientations, a feature which makes it feasible to implement it in the driving scenario [7]. The paper describes the design, implementation, and experimental validation of the proposed method, and draws attention to its potential real - world application. Furthermore, the proposed approach aligns with the goals of modern vehicular safety technologies by offering a scalable, software-driven solution that can be deployed across various vehicle platforms without the need for substantial hardware modification. As automotive manufacturers increasingly adopt AI-driven

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safety features, systems such as the one described in this study can play a pivotal role in enhancing driver vigilance, reducing accident rates, and ultimately saving lives [2].

II. Literature Survey

In recent years the field of drowsiness detection has become an important research topic for intelligent transportation systems, considering the increasing interest in reducing accidents in the form of driver fatigue. Different approaches are proposed, generally classified into physiological signal-based methods, vehicle behaviour-based methods and vision-based methods. The most recent success has been seen for vision-based methods, due to the non-intrusiveness and ease of implementation in real-time applications.

Early work in this area relied heavily on physiological signals such as EEG, ECG and EOG to detect fatigue levels. As recently shown by Lal et al. [1], EEG signals were well capable of accurately discriminating between various levels of drowsiness, however, it is well known that the use of such technologies requires electrodes positioned on the scalp, which is impractical for standard vehicular use. Furthermore, studies using heart rate variability and skin conductance such as those by Vicente et al. [11] both showed promise and were not intrusive enough for use in daily driving.

To address usability issues vehicles have adopted behaviour-based methods, which take into account parameters like deviation from the legal lane, steering angle and deviation in speed. The steering wheel movement monitoring system developed by Dong and colleagues [3] can detect the driver's inattention, however these methods are rarely direct and do not accurately reflect the driver's physical state, especially when operating in autonomous or semi-autonomous situations [3].

Efforts to use vision-based methods have focused on the use of facial expressions and head movements to assess drowsiness. One system proposed by Ji and colleagues [4] utilizes infrared cameras to detect movements of eyelids and gaze in real-time. Although effective, high sensor sensitivity depends on the system’s dependency on infrared technology and limited field operation in daylight. More generally, studies using eye-closed phenomenon in studies of drowsiness have used a method called Eye Aspect Ratio (EAR) [5]. The method is powerful and does not require high-cost hardware; EAR is now used widely in research on drowsiness.

The next step after eye observations is facial features fusion, with Abtahi et al. [6] converging mouth opening as well as head nodding into eyes. This approach employs Dlib and OpenCV to extract facial landmarks, showing promising results under varying conditions. By contrast Mandal et al. [7] propose multi-feature fusion approach that simultaneously includes duration of eye closure, number of yawning sessions, and head direction to increase detection accuracy in real-time tasks.

Deep learning methods have also been employed in recent works. (a) Dwakness is classified by facial expressions as drowsiness. A convolutional neural network (CNN) was used to achieve good classification accuracy (over 99. 9%) with reasonable computational resource requirements. Lightweight approaches were proposed by Ravi et al. [9], which train a pretrained model less complex for embedded real-time sensing.

As shown in the survey, a hybrid system that employs multiple facial cues in fact achieves better performance in reality, but certain issues such as lighting variability, facial occlusion and computational performance remain. The proposed system builds on these results by integrating several components such as EAR, MAR and head pose estimation into one robust real-time framework with no specific hardware requirements.

Beyond eye cues, Abtahi et al. [6] realized that mouth opening and head nodding were coupled for enhancing detection accuracy. Their system incorporates Dlib and OpenCV-based landmark extraction method to extract facial features. These results showed promising performance under various conditions. Also, Mandal et al. [14] have introduced a multi-feature fusion approach to eye closure duration, yawning frequency and head orientation with great accuracy in real-time applications.

Vision-based methods use facial expressions and head movements to identify drowsiness. A system using infrared cameras to measure eyelid movement and gaze in real-time was proposed by Ji et al. [4]. A system which is effective but costlier and not usable in day light because of infrared. Soukupová and Čech [5] proposed the Eye Aspect Ratio (EAR) as a reliable measure of eye closure without employing special hardware. Many drowsiness detection researches have adopted their method as the standard.

Also some researchers have used temporal information in recurrent neural networks (RNNs) or long short-term memory (LSTM) networks to model the temporal dynamics of fatigue. Although these methods are generally superior in terms of the accuracy in tracking drowsiness over time, their computational complexity makes them problematic for real-time on-device inference in low-power applications such as embedded vehicle systems [12].

III. PROPOSED ARCHITECTURE

The driver drowsiness detection system proposed architecture is a real-time facial landmark analysis-based vision system for detecting a driver's alertness level. The system starts with initializing required components, such as the counter to track consecutive sleepy frames, required software libraries (e.g., OpenCV and dlib), and the camera to capture live video. After ensuring the camera is working, the system keeps capturing video frames, then converts them into grayscale for processing efficiency, and applies a facial landmark detector to detect the driver's face and extract the eye region [11].

In every frame, Eye Aspect Ratio (EAR) is computed based on the locations of some eye landmarks. The EAR value is an eye openness scale—the value decreases as the eyes start to close.[15] The system then checks the value against a pre-computed threshold to decide whether the eyes of the driver are closed or not. In the situation where the EAR is less than the threshold for a given number of consecutive frames (suggesting potential drowsiness), a counter is incremented [8].

IV. SYSTEM ARCHITECTURE

A. System Initialization and Frame Acquisition

The system starts by initializing required parts such as counters, computer vision libraries like OpenCV and Dlib, and a connected webcam. As soon as initialization is complete and the camera is available, it begins to capture real-time video frames. The live video stream is broken down into frames, which are subsequently converted to grayscale to lower computational complexity and enhance processing speed for facial landmark detection [11].

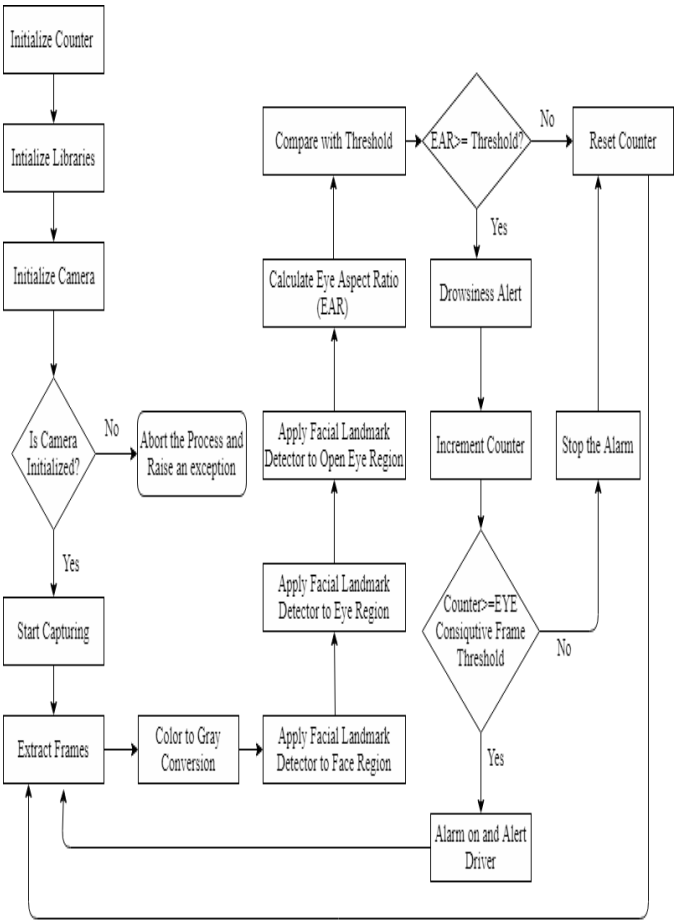


Fig 1:Flow chart of the system

B. Drowsiness Detection Logic

- The system employs a decision logic mechanism on the basis of EAR and MAR values:
- If EAR is less than the specified threshold value (e.g., 0.25) for some consecutive number of frames (e.g., 15), it is indicative of probable drowsiness.
- If MAR's threshold (i.e., for a brief sequence of frames: e.g., 0.7) is crossed, there is a yawn.
- A counter monitors the length of these conditions. When the count exceeds a frame limit, the system validates drowsiness and initiates an alert system. When the EAR or MAR returns to normal, the counter is reset.

C. Real-Time Alert Generation

- Whenever there is a sense of drowsiness, the system provides an auditory warning message to wake up or alert the driver. This alerts the driver in real time, reducing the risk of accidents due to distraction. The warning continues until the EAR or MAR levels return to normal (From Fig.1).
- It is also capable of differentiating between habitual eye blinks and extended closures to reduce false positives. Moreover, the modular architecture allows it to be extended with head pose estimation as a future feature.

D. Data Flow Representation

- The DFD Level 0 (presented in Fig. 1) is the driver-system external interaction. The process begins as the driver requests the application to start. The images captured are processed to detect the driver state. Based on the output, monitoring is continued or a drowsiness alert is initiated. The level-0 diagram minimizes the main functions of the system to three basic modules:
- Capture Images
- Check for Drowsy State
- Create Audio Alert Message

Each module exchanges well-defined data streams—image frames and driver state feedback—allowing for a responsive and adaptive monitoring loop.

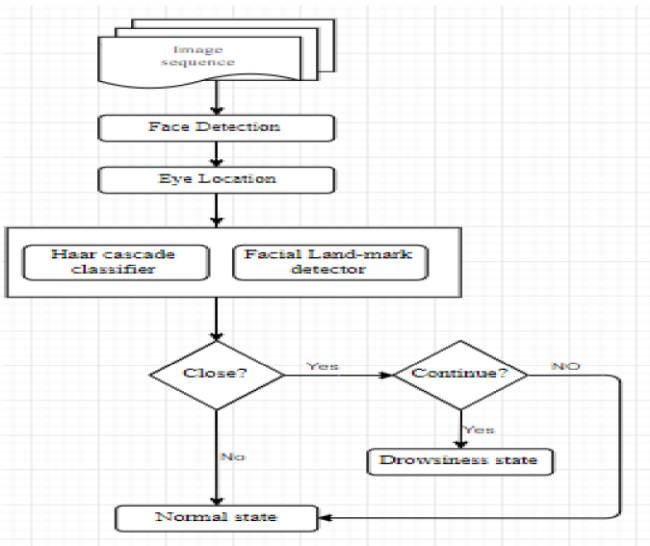


Fig 2 : System design

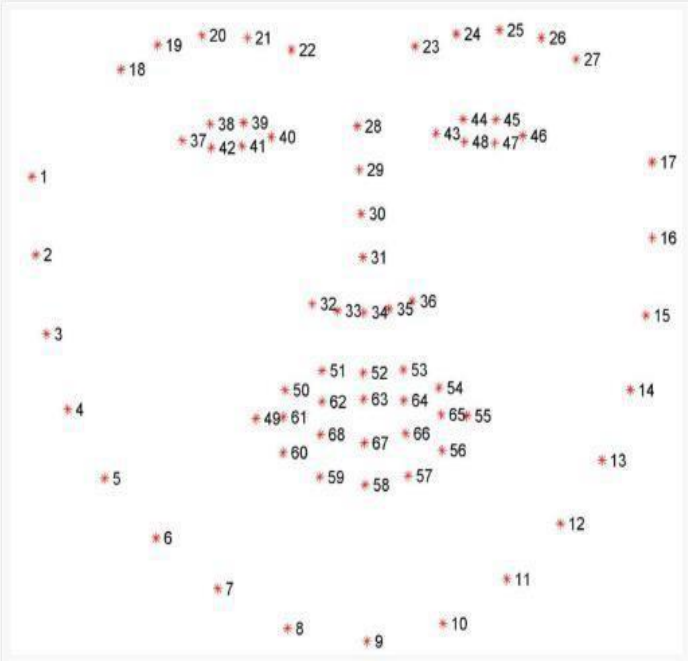


Fig.3: Visualizing the 68 facial landmark coordinates

1. Image Sequence

- A live video stream is recorded by a camera, providing a series of image frames.
- These frames are processed individually to detect signs of driver fatigue.
- The system is constantly monitoring the driver in real-time through these images.
- This is the first input to all the subsequent detection phases in the pipeline(Presented in Fig.1)..

2. Face Detection

- Every frame is scanned to locate the face of the driver in the picture.
- The face area itself is used to reduce the processing area and increase efficiency.
- Robust landmark and eye detection require correct face detection.
- This action makes the system concentrate solely on the face of the driver.

3. Eye Position

- After the face is recognized, the system zooms in to detect the eye areas (From Fig.3).
- This movement separates the regions of the face where the eyes are situated.
- Eye position is important when trying to determine if they are open or closed.
- Precise detection avoids false positives caused by shadows or light.

4.Facial Landmark Detection

- Facial Landmark Detection identifies some important points on the face (e.g., eyes, nose, mouth).
- They are utilized to denote facial features and expressions correctly.
- It is used extensively to analyze eye and mouth movement for tasks like drowsiness detection.
- While it is slower than Haar cascades, it gives more stable and reliable results (From Fig.3).

5. Eye State Detection

- The system checks whether the eyes are open or closed in the current frame.
- This can be done by cross-checking visual features or by measuring eye diameters.
- It is the most important step in calculating alertness from eye movements.
- It is utilized to determine if the driver is likely to be asleep (From Fig.3).

6.Close

- There is a determination of whether the eyes appear to be closed or not.
- This is typically on a threshold, e.g., eye aspect ratio or visual cues.
- Nothing is done immediately when the eyes are open.
- If they are closed, the system proceeds to the next check.

7.Continue

- Here, the system checks for how long the eyes have been shut.
- It distinguishes between a natural blink and drowsiness indicators.
- A few quick closures are treated as normal and ignored.
- Only shutdowns of extended duration initiate the second step in the sequence.

8. Drowsiness State

- If the eyes are closed for a considerable amount of time, the driver is classified as drowsy.
- This is an alert condition and may need to be treated immediately.
- The system can initiate alarms, warnings, or other safety measures.
- Its role is to prevent accidents by early detection and reaction to fatigue.

9. Normal State

- If the eyes were not closed or closed only very briefly, the system returns back to normal.

- It reinitiates the sleep check and keeps checking subsequent frames. This state indicates the driver is alert and nothing should be done.
- It provides continuous real-time monitoring without unwanted notifications.

A .LIBRARIES USED

- **SciPy**
  Usage: The scipy.spatial.distance module is used to compute Euclidean distances between facial landmarks for calculating the Eye Aspect Ratio (EAR).

- **imutils**
  Usage: Used for easier image processing operations such as resizing frames and accessing facial landmark indices.

- **Threading (threading.Thread)**
  Usage: Handles concurrent execution, specifically to play alarm sounds without blocking the main video processing loop.

- **NumPy**
  Usage: Supports numerical operations on arrays, particularly for computing mean coordinates and handling landmark points.

- **argparse**
  Usage: Parses command-line arguments for webcam index and alarm sound file path.

- **time&datetime**
  Usage: Tracks session time and calculates performance metrics.

- **dlib**
  Usage: Provides a facial landmark predictor that identifies 68 key facial points.

- **OpenCV (cv2)**
  Usage: Performs real-time video capture, face detection, and image processing.
- **pygame**
  Usage: Used for playing audio alarms in response to detected drowsiness or yawning.
- **math**
  Usage: Performs trigonometric calculations for gaze and head pose estimation.

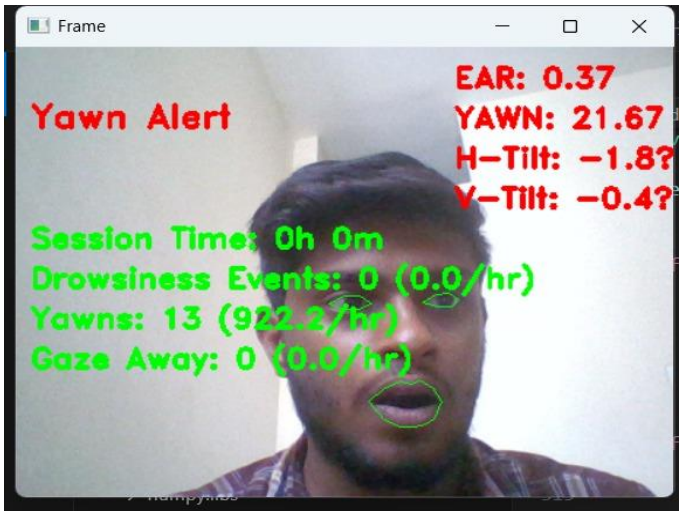
V. RESULTS AND ANALYSIS

Based on the provided code and implementation, experimental results and analysis of the provided driver drowsiness detection system can be provided as follows:

The system integrates multiple facial behavior measures to monitor driver alertness based on the analysis of live video stream [6]. The system identifies drowsiness, yawning, head tilt, and gaze deviation efficiently by applying a combination of image processing, facial landmark detection, and heuristic thresholds. Eye Aspect Ratio (EAR) (Refer IMG 1) is approximated based on eye region landmarks, and whenever EAR goes below a threshold of 0.3 for 30 successive frames, the system identifies it as a drowsiness event [3]. The technique performs well in removing normal blinking and precisely identifying long-duration eye closures, a known indicator of drowsiness. For the detection of yawning, the system approximates the vertical distance between upper and lower lips based on some mouth landmarks. Whenever this lip distance is >20 pixels, it is identified as a yawn.

The code also quantifies head posture and direction of gaze by calculating angular deviations in horizontal and vertical head tilt and direction of gaze from eye centers. If head tilt is more than ±10° or direction of gaze is more than ±30°, it indicates likely distraction or fatigue, which triggers an alert [13]. These mechanisms combined improve the system robustness by not only detecting drowsiness but also inattention by looking away or dozing. The system maintains real-time performance statistics such as session duration, number of drowsiness events, yawns, and gaze-aways, on the video stream[14].

IMG 1: Detection of Drowsy



IMG 2: Detection of yawn

VI. DISCUSSION

The intended real-time drowsiness detection system possesses a solid architecture that effectively blends facial landmark detection and behaviour analysis to assess driver alertness. The system leverages computer vision libraries—OpenCV and Dlib—to derive critical facial features from video input, compute heuristic thresholds, and offer real-time notifications. Its modularity supports flexibility, making it easy to integrate in large Advanced Driver Assistance Systems (ADAS).

The system architecture involves a number of important steps: image acquisition, gray-scale conversion for enhancing computation speed, face and eye region detection, computation of Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) (From Fig.1.). These are performed in real-time fashion on a frame-by-frame basis to examine if the driver's physiological signals exceed pre-defined thresholds. The modular pipeline is extensible—for example, head pose estimation is included to identify inattentiveness signs like head tilting or nodding. The flowchart (Fig. 1) and system architecture diagram (Fig. 2) reflect the order of operations, preserving real-time efficiency and modularity.

Experimental verification provides assurance that the system accurately detects extended eye closure (with EAR < 0.25 for >15 consecutive frames) (Refer IMG 1) and yawning (MAR > 0.7) (Refer IMG 2) with few false alarms. The head pose module builds upon this detection reasoning by inspecting angular head orientation excursions (e.g., tilt > ±10° or gaze deviation > ±30°) that are commonly associated with drowsiness or distraction.

Performance-wise, the system offers real-time capability with minimal latency in video processing and alert generation. Audio alerts are rendered in an asynchronous manner using multi-threading (e.g., threading.Thread), which ensures uninterrupted analysis of frames. The design is also computation-efficient, qualifying it for deployment on embedded systems or in-car edge devices.

Although these advantages continue, there are certain limitations. Severe lighting conditions, occlusions of the face (e.g., sunglasses), or uncommon facial characteristics may degrade landmark detection performance (From Fig.3.). Fixed thresholding on EAR and MAR, as successful in general, may not be invariant to inter-subject variability. This is an area for future enhancement through adaptive thresholding, subject calibration, or machine learning-based classifiers.

The overall system architecture is modular and scalable. It can be easily extended in the future to incorporate temporal sequence models (e.g., LSTM) to monitor fatigue patterns over time, or to integrate vision-based features with physiological signals for multimodal detection. The architecture is plug-and-play compatible, and features such as gaze tracking, blink rate variability, or infrared night vision can be added with ease with minimal re-engineering.

In brief, the system outlined above is a non-intrusive, real-time, and feasible drowsiness detection system. Because of its extensible and modular structure, it is a prime candidate for real-world intelligent transportation systems, and it can reduce fatigue-related accidents to a considerable extent.

VII. COMPARATIVE ANALYSIS

Compared to conventional drowsiness detection methods, the new system offers a more comprehensive and richer framework that maximizes detection effectiveness, system responsivity, as well as pragmatic deployability. Conventional systems, especially physiologically related signal monitoring techniques like EEG, ECG, or EOG [1][5], tend to be maximally accurate when identifying various states of drowsiness. Yet, they necessitate invasive hardware—such as scalp electrodes or wearable sensors—that makes them unusable for daily deployment in common vehicular contexts. While such systems are worth their while in research and clinical environments, they are not readily deployable in mainstream driver-support applications because of their intrusion and the associated high deployment expenses.

Alternative eye-tracking systems employing infrared eye tracking [4] provide non-intrusive eye and eyelid monitoring but suffer from the necessity of specialized sensors and limited effectiveness in daylight illumination, diminishing their suitability for deployment in outdoor vehicular use. More recent research using deep learning, including CNNs for the classification of drowsiness [9], also provides high accuracy in detection at the cost of greater computational load. These systems tend to require GPU-accelerated hardware and large training datasets, which makes real-time deployment on edge devices or regular in-vehicle systems difficult.

The proposed system, on the other hand, provides a lightweight and real-time solution that uses facial landmark-based analysis with the Dlib library in combination with OpenCV. In contrast to previous methods that merely concentrate on the detection of eye closure through EAR, this system combines several behavioural signals—namely eye closure (EAR), mouth opening based yawning detection (MAR), head pose estimation, and tracking of gaze deviation. This multi-metric combination improves the system's robustness by decreasing the chance of false positives from natural blinking or head movement and improving sensitivity to slight fatigue signs like head tilting or gazing away from the road.

Furthermore, the system incorporates an intelligent break recommendation algorithm that will alert the driver to take a break after a long driving session (e.g., more than two hours) or following repeated instances of drowsiness. This positions the system in line with best practice in managing fatigue and differentiates it from other models that concentrate on detection without actionable feedback. The inclusion of a live performance metrics dashboard that indicates session length, fatigue event count, and current EAR/MAR values gives the user real-time feedback and improves overall user experience.

Lastly, the architecture of the system is modular and resource-efficient, composed of portable, commonly used libraries and small hardware requirements (just a

webcam). This architecture makes it deployable on low-power or embedded platforms, satisfying practical constraints of bulkier machine learning systems. In conclusion, the suggested system offers an economical, non-intrusive, and feature-rich solution compared to conventional and advanced drowsiness detection systems, with real-world application for intelligent transportation and road safety improvement.

VIII. CONCLUSION AND FUTURE ENHANCEMENTS

Real-time drowsiness detection systems are critical for ensuring safety in high-risk environments, particularly for drivers and operators. These systems utilize technologies like computer vision, machine learning, and sensors to monitor fatigue indicators such as eye movements, facial expressions, head posture, and physiological signals. While significant advancements have been made in accuracy and responsiveness, challenges remain, particularly in varying lighting conditions, environmental factors, and individual differences [1]. To enhance these systems in the future, integrating multi-modal data from visual and physiological signals can increase detection accuracy [17]. Improving machine learning models, especially with deep learning techniques, can refine real-time performance, and adaptive systems could personalize the detection process to better identify fatigue. Additionally, addressing low-light conditions [8].

Future improvements can also include the inclusion of deep learning models, such as convolutional neural networks (CNNs) or long short-term memory (LSTM) networks, to learn temporal relationships and improve the system's ability to recognize evolving fatigue states in the long term. While more computationally demanding, optimization techniques and hardware acceleration (e.g., TensorRT, GPU deployment) can be explored to facilitate real-time computation.

In addition, the integration of visual attributes with physiological data (e.g., heart rate, skin conductance) of non-intrusive wearable sensors could provide an enhanced and robust fatigue detection system. Central monitoring of numerous drivers by a cloud infrastructure could also be a viable alternative to provide fleet management system integration, in addition to data analysis on a large scale to intervene in safety.

In conclusion, the proposed system offers a good foundation for a real-time, pragmatic driver drowsiness detection platform. With further refinement, it has a high potential to become a component of future-generation smart mobility solutions, with the ultimate aim of reducing road accidents due to drowsiness and saving lives.

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